SPACE FACULTY

# Let’s learn about **satellites**!

**Space mission** and **satellite** design course

# Payloads and Satellite Missions

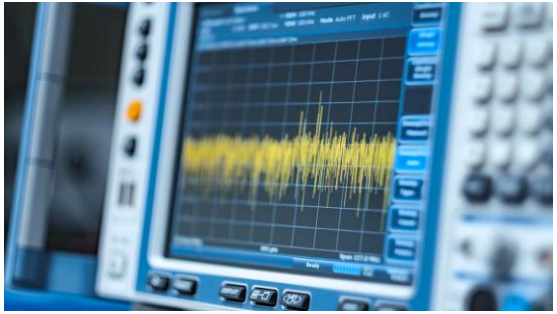
2NDSpace

# Early Cubesat Missions

Early CubeSats were primarily developed for **education** and **technology demonstration**.

The first missions aimed to prove that very small satellites could achieve meaningful objectives despite tight constraints on volume, mass, power, and communications.

Typical goals included hands-on training for students, in-orbit validation of new technologies, and simple scientific experiments.

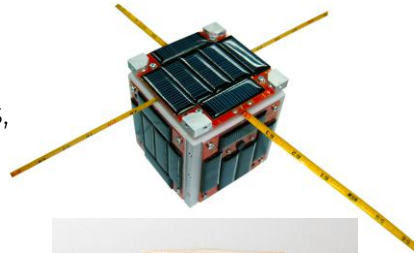


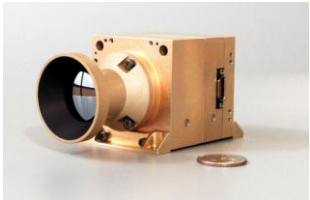
SPACE FACULTY
2NDSpace

# Early Cubesat Missions

Basic payloads in early CubeSat missions typically fell into three categories: communications, imaging, and data collection.

- **Communication**: To demonstrate basic telemetry and command links, often using amateur radio bands. Bandwidth and data rates were usually very limited.
- **Imaging**: Simple cameras were used for Earth observation and educational demonstrations, with modest resolution compared to today's standards.
- **Data Collection**: Basic sensors enabled small scientific experiments and in-orbit measurements, constrained by the tight limits on volume, power, and downlink capacity.





SPACE FACULTY
2NDSpace

# Challenges Faced with Early CubeSat Technology

Early CubeSats were strongly constrained by miniaturization, limited power availability, and restricted communication capability. These constraints directly affected payload performance and reliability.

Integrating functional components into extremely small volumes created significant engineering challenges. Limited electrical power reduced operational margins, while low data rates and link instability often compromised mission objectives.

Statistics show that out of 270 CubeSats launched between 2002 and 2016, 139 (51%) experienced mission failures. Many of these failures were related to power system issues, communication problems, or insufficient payload robustness.

These results highlight how critical careful payload design and system integration are for mission success, especially within the tight constraints of CubeSat platforms.

SPACE FACULTY
2NDSpace

# Modern CubeSat Missions

Modern CubeSats have evolved far beyond educational demonstrations and now support a wide range of scientific and commercial applications.

Typical mission areas include:

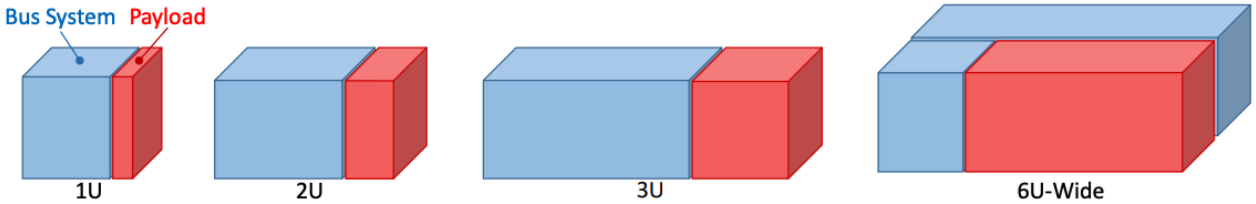
- Earth observation and remote sensing.
- Ionospheric and upper-atmosphere research.
- Radiation and space environment monitoring.
- Space weather observation.
- Advanced educational and technology demonstration missions.
- Deep-space and interplanetary feasibility studies.
- Solar particle and heliophysics observation.
- Asteroid exploration and prospecting concepts.
- Maritime and traffic surveillance.

SPACE FACULTY
2NDSpace

# CubeSat Missions: Payload and Bus System

The mission's capability is influenced by the CubeSat's dimensions. The missions, key considerations include:

- **Optimized Bus Systems**: Larger CubeSats can host more advanced subsystems improving overall mission performance.
- **Increased Power and Energy Storage**: More surface area enables larger solar arrays, while additional volume allows higher-capacity batteries, supporting more demanding payloads.
- **Enhanced Processing and Communications**: Greater internal space accommodates more powerful onboard computers, expanded memory, and higher-data-rate communication systems.
- **Improved Attitude Determination and Control**: Larger platforms facilitate integration of precise ADCS components (reaction wheels, star trackers), enabling higher pointing accuracy.
- **Accommodation of Larger Payloads**: Increased volume and mass budgets allow integration of more complex and higher-performance scientific instruments.




SPACE FACULTY
2NDSpace

# Payloads

In the context of space missions, the payload refers to the set of instruments or equipment carried by a spacecraft to accomplish its mission objectives.

It includes all components that directly generate mission value, such as scientific instruments, sensors, cameras, communication devices, or experimental hardware.

The nature of the payload depends entirely on the mission goals and defines what the satellite is ultimately designed to do in orbit.

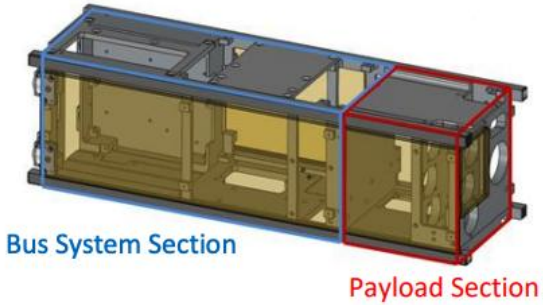
SPACE FACULTY
2NDSpace

# Payloads

For CubeSats and small satellites, the payload is usually the central element of the mission. These platforms are often designed for a specific scientific, educational, or technological objective.

Understanding the payload means understanding the mission itself: the type of data to be collected, the experiment to be performed, or the service to be delivered.

For this reason, the entire satellite architecture, including the bus subsystems, is designed around the payload requirements.




SPACE FACULTY
2NDSpace

# Typical Payloads

Cubesats can host a diverse range of payloads:

- **Optical Payloads**: Optical and multispectral imaging cameras.
- **Communication Payloads**: Radio transponders, software-defined radios.
- **Scientific Experiment Payloads**: Instruments for space research.
- **In-Orbit Demonstration (IOD) Payloads**: Components or technologies validation.
- **Internet of Things (IoT) Payloads**: Contributions to global IoT networks.
- **Software Payloads**: Experimental algorithms or software technologies.

SPACE FACULTY
2NDSpace

# Optical Payloads

Optical payloads in CubeSats include a range of instruments designed to capture and analyze electromagnetic radiation, enabling applications from Earth observation to astronomy.

Imaging payloads are the most common type:

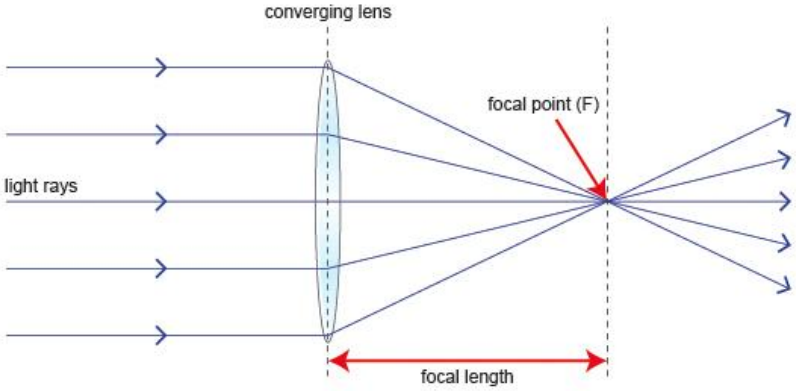
- **Visible Cameras**: Compact optical systems operating in the visible spectrum, typically used for Earth imaging, monitoring specific regions, or observing celestial objects.
- **Multispectral and Hyperspectral Cameras**: Instruments that acquire data across multiple spectral bands. Multispectral systems capture a limited number of discrete bands, while hyperspectral systems measure hundreds of narrow bands, enabling detailed analysis of vegetation, water quality, minerals, and surface composition.

SPACE FACULTY
2NDSpace

# Visible Light Camera

A digital camera works by capturing and converting light into digital signals that can be processed and stored.

Light enters the camera through the lens, which focuses the light onto the camera's image sensor. The image sensor is a semiconductor device that consists of millions of tiny light-sensitive elements called **pixels**.



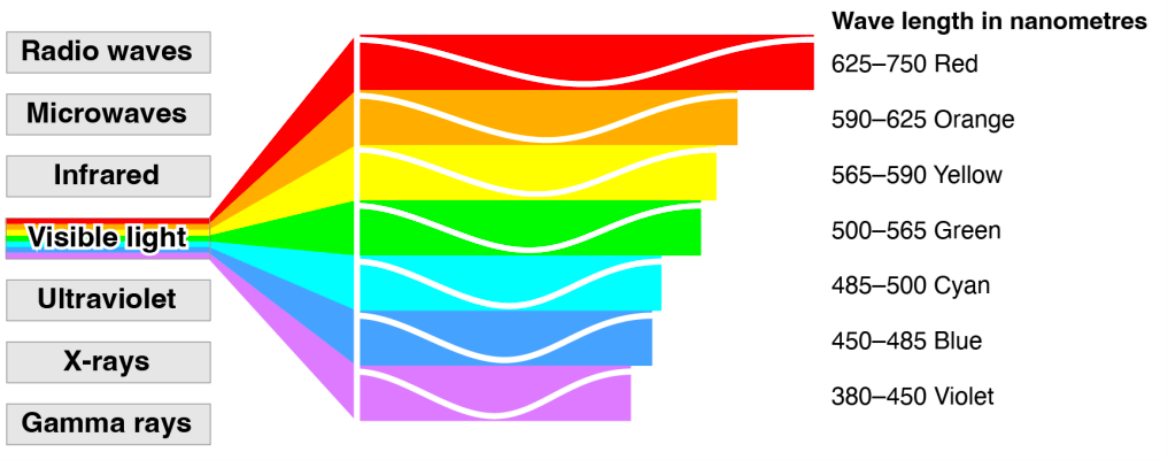

SPACE FACULTY
2NDSpace

# Visible Light Camera

Each pixel on the image sensor has a light-sensitive material (often a photosensitive diode or a photosite in a charge-coupled device, CCD, or a complementary metal-oxide-semiconductor, CMOS, sensor).

# Visible Light Camera

When light strikes a pixel, it generates an electric charge in proportion to the intensity and color of the light.




The accumulated electric charge in each pixel is then converted into a digital signal through a process called analog-to-digital conversion (ADC).

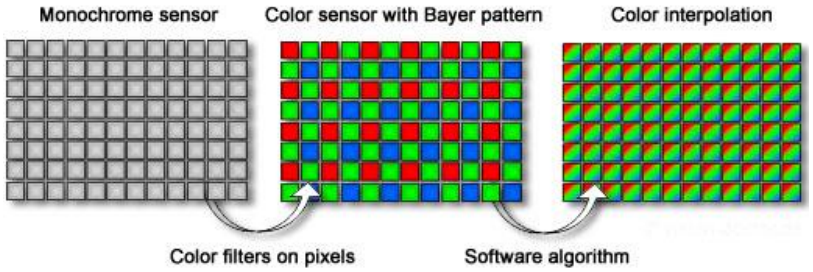
The ADC assigns a digital value to each pixel based on the amount of light it received.

SPACE FACULTY
2NDSpace

# Visible Light Camera

In color cameras, each pixel is typically overlaid with a color filter, allowing it to capture information about the intensity of red, green, and blue light.

The camera combines the information from neighboring pixels to reconstruct the full color of each point in the image.




SPACE FACULTY
2NDSpace

# Visible Light Camera

The digital signals from the image sensor are processed by the camera's onboard electronics, including a Digital Signal Processor (DSP).

The DSP performs various tasks such as white balance adjustment, noise reduction, and sharpening to enhance the quality of the image, via appropriate filters.

Filters serve multiple purposes:

- **Noise reduction**: Filtering techniques, such as Gaussian smoothing or median filtering, remove random variations and disturbances in pixel values caused by sensor imperfections or transmission errors.

Original Image (Left) Vs. Gaussian Filtered Image (Right)

# Visible Light Camera

- **Edge detection**: Filters like the Sobel or Laplacian filter emphasize rapid intensity changes (edges and boundaries), aiding in the identification of important features and structures.

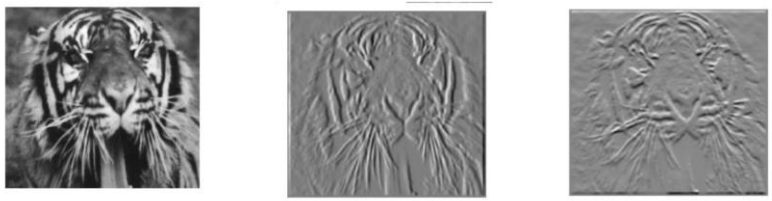
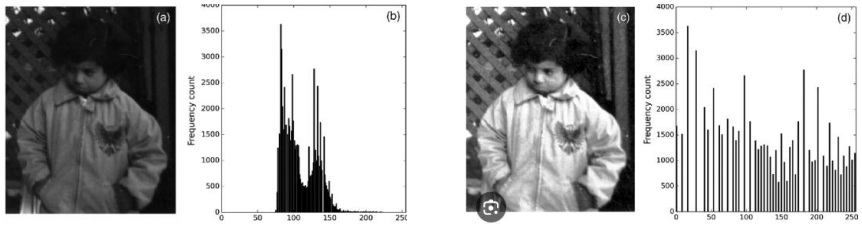


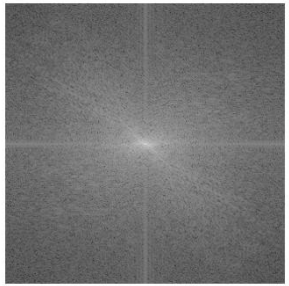
Image $f$ $\qquad$ $\dfrac{\partial f(x,y)}{\partial x}$ $\qquad$ $\dfrac{\partial f(x,y)}{\partial y}$

- **Enhancement of image features**: Techniques like contrast stretching, histogram equalization, or the application of specific frequency filters help bring out certain features.



# Visible Light Camera

- **Frequency analysis**: Transformations like the Fourier Transform allow the analysis of an image's frequency components, helping to identify patterns, textures, and structures.



- **Image compression**: Filters may be used to identify and retain essential image information, minimizing redundant or less critical data during compression.

**Figure 2:** Reconstructed image of previous method and present work for HP values=10,20,30,40,50,60 shown from top left

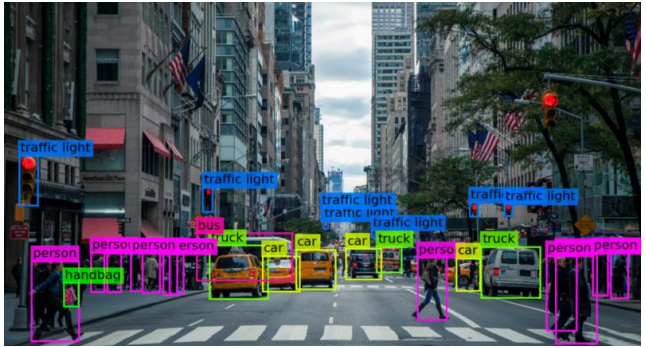
SPACE FACULTY
2NDSpace

# Image Filtering

Filters play a crucial role in computer vision by enhancing or modifying the visual information contained in images.

**Computer vision** is a multidisciplinary field of AI that empowers machines to interpret and make decisions based on visual data. It aims to enable computers to mimic human vision by recognizing patterns, extracting meaningful information, and understanding the content of images or videos.

Computer vision encompasses a wide range of tasks, including image recognition, object detection, segmentation, facial recognition, and scene understanding.




2NDSpace

# Computer Vision: An Example

Let's have a look at the following image of LEGO bricks:

How do we create an algorithm that is capable of recognizing, counting and classifying the bricks from a picture taken by the camera?

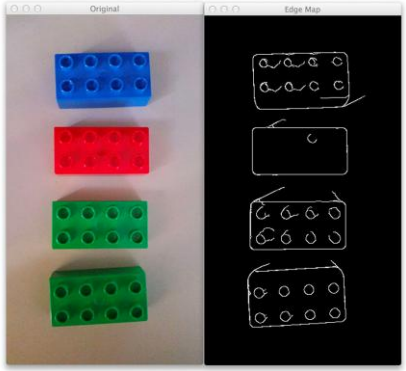
Question: *Which features would you look for to do so?*

SPRICE FACULTY
2NDSpace

# Computer Vision: An Example

Recognizing, counting, and classifying a set of LEGO bricks in an image involves several steps.
A simplified example using common computer vision techniques:

- **Image Acquisition**: Begin by obtaining an image containing LEGOs.
- **Preprocessing**: Apply preprocessing techniques to enhance the quality of the image.
- **Color Segmentation**: LEGO bricks often have distinct colors. Apply color segmentation to separate the image into regions corresponding to different colors.
- **Edge Detection**: Use edge detection filters, to identify the boundaries of individual LEGO bricks.




- ...

- **Classification**: Train a machine learning model, such as a simple classifier or a more sophisticated neural network, using the extracted features.

- **Classifying Blocks**: Apply the trained model to classify each remaining contour as a specific type of LEGOs (e.g., brick, plate, or tile). This step enables the algorithm to recognize and label each block in the image.

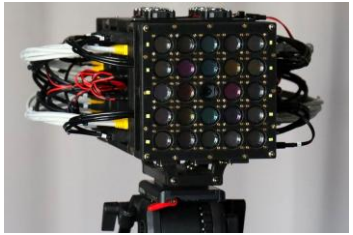
SPRICE FACULTY
2NDSpace

# Computer Vision and Camera Payload

A CubeSat camera payload can include a dedicated onboard processor, often implemented as a separate **Payload On-Board Computer** (Payload OBC), enabling real-time image processing and AI-based analysis.

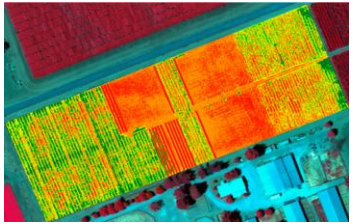
This architecture allows the payload to operate as a semi-independent subsystem, performing onboard data processing, feature extraction, or event detection before transmission. As a result, the amount of data to be downlinked is reduced, the **main platform OBC** is relieved from computationally intensive tasks, and overall mission efficiency is significantly improved, especially in Earth observation missions.

# Multispectral & Hyperspectral Cameras



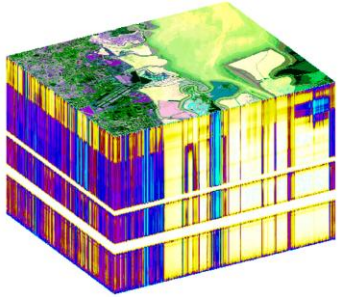
**Multispectral cameras**

- Capture a small number of discrete wavelength bands (typically ~3–10), instead of only RGB.
- Used for Earth observation applications such as agriculture, environmental monitoring, and land/vegetation classification.
- Usually offer moderate to high spatial resolution (higher spatial resolution = finer ground detail).



**Hyperspectral cameras**

- Capture hundreds of very narrow, contiguous bands across a continuous spectral range.
- Enable material discrimination (subtle differences in composition), useful in environmental monitoring, geology/mineral mapping, and some security/defense use cases.
- Generate much larger datasets and require more complex processing.



SPACE FACULTY
2NDSpace

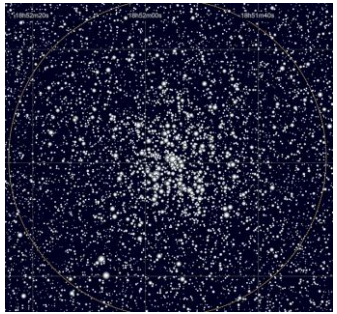
# Optical Payloads

**Star Trackers**:

A star tracker is an optical attitude sensor that determines a spacecraft's orientation by imaging the star field and identifying stars.

Using an onboard star catalog, the tracker detects star patterns within its field of view and matches them to known constellations.

From this match, onboard processing computes the spacecraft's 3-axis attitude and updates it continuously as new images are acquired.



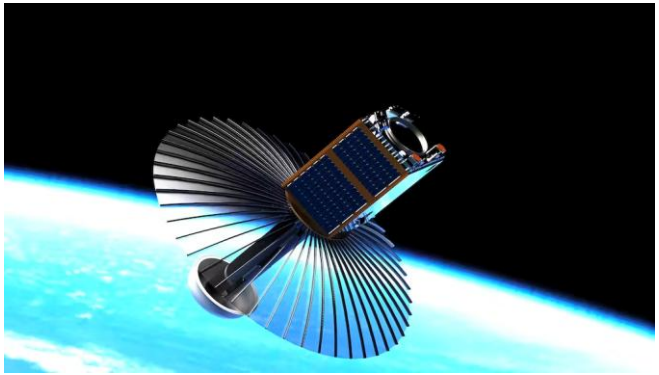
# Optical Payloads

**Synthetic Aperture Radar (SAR)**:

SAR is an active remote sensing system that transmits microwave signals toward the Earth and records the reflected echoes.

By exploiting the motion of the satellite, it synthesizes a large “virtual” antenna aperture, achieving high resolution in both range and azimuth.

SAR provides detailed information on surface structure, topography, vegetation, and environmental changes, and operates in all weather conditions, independently of sunlight (day and night imaging).



SPACE FACULTY
2NDSpace

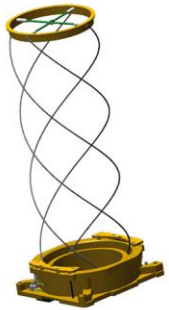
# Communication Payloads

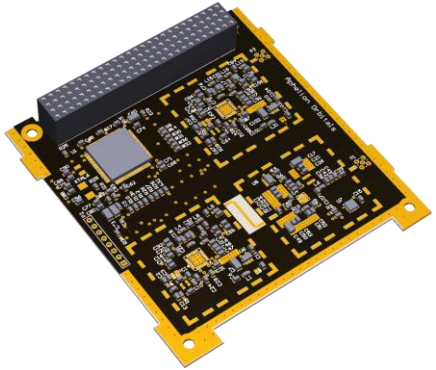
Common types of communication payloads found on cubesats:

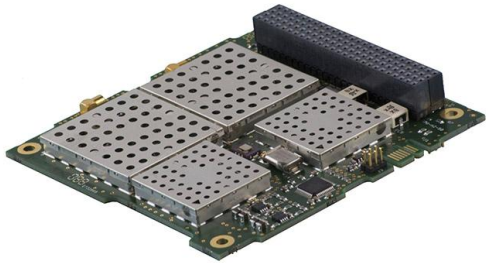
- **UHF/VHF Transceivers (300 MHz to 3 GHz)**: Ultra High Frequency (UHF) and Very High Frequency (VHF) transceivers are commonly used for communication with ground stations. They operate in the amateur radio bands and allow cubesats to establish two-way communication.
- **S-Band Transceivers (30 MHz to 300 MHz)**: S-Band transceivers operate in the microwave frequency range and are used for higher data rates. They provide better communication capabilities compared to UHF/VHF, making them suitable for missions with higher data transfer requirements.
- **X-Band Transceivers**: X-Band is another microwave frequency band used for communication in cubesats. It offers higher data rates than S-Band and is suitable for missions that demand significant amounts of data to be transmitted.
- **Software-Defined Radios (SDRs)**: SDRs provide flexibility in terms of communication frequencies and modulation schemes. They allow for reprogramming the communication system onboard, enabling adaptation to different mission requirements.
- **Antennas**: Various antenna configurations are used depending on the mission and frequency band. Deployable antennas, such as tape or whip antennas, can be used to increase the effective communication range.

SPACE FACULTY
2NDSpace

# Communication Payloads







SPACE FACULTY
2NDSpace

# Optical Communication

**Optical communication** for CubeSats uses a laser beam to transmit data through free space. Compared to RF, it can deliver much higher data rates with low mass and power, making it attractive for small satellites.

Main elements include:

- Laser transmitter: generates a modulated optical signal to encode data.
- Pointing optics: collimates and directs the beam to achieve the required link budget and pointing accuracy.
- Modulation and protocol: defines how data is encoded and synchronized on the optical carrier.
- Safety and compliance: operational constraints and procedures reduce the risk of interference and meet applicable laser safety and regulatory requirements.

SPACE FACULTY
2NDSpace

# Scientific Experiment Payloads

CubeSats can host specialized instruments or sensors designed to perform scientific measurements, observations, or in-orbit experiments. These payloads collect mission-specific data, environmental parameters, particles, radiation levels, or other physical phenomena.

# In-Orbit Demonstration (IOD) of Subsystems

In CubeSats, **In-Orbit Demonstration (IOD)** missions leverage the fact that almost any subsystem can act as a payload.

This means that not only dedicated instruments, but also individual components, such as a new communication module, a novel sensor, or an advanced power or processing unit, can be flown specifically to validate their performance in space.

IOD missions highlight the flexibility of CubeSats and provide a cost-effective way to raise the **TRL** of new technologies before large-scale implementation.

Example: a 3U satellite flying an experimental OBC as a payload, in addition to the platform's main OBC, specifically to demonstrate and validate it in orbit.

SPACE FACULTY
2NDSpace

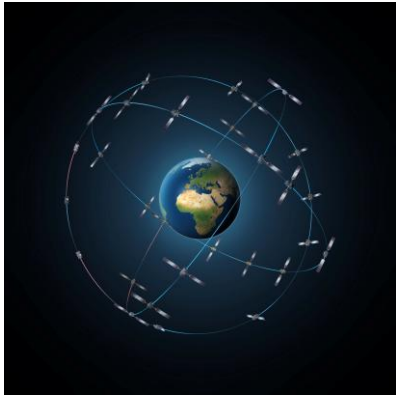
# Internet of Things (IoT) Payloads

CubeSats can host **IoT payloads** that connect remote sensors to the internet, enabling near real-time data collection and delivery from anywhere on Earth.

By flying in LEO constellations, these missions provide global coverage and frequent revisits, which is ideal for monitoring assets and environments in hard-to-reach areas.

Typical applications include agriculture, environmental monitoring, disaster response, and wildlife tracking, where CubeSats collect sensor data and forward it to ground networks.

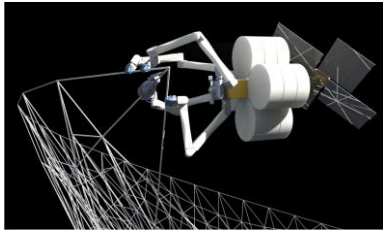
In more advanced architectures, satellites can share data via inter-satellite links and perform basic onboard processing or data fusion to reduce downlink needs and improve responsiveness.



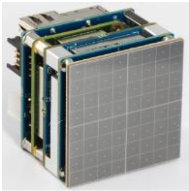
SPACE FACULTY
2NDSpace

# Advanced technologies

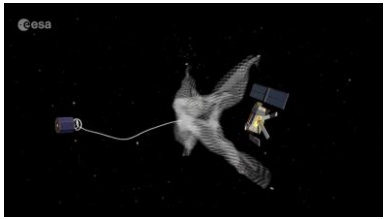
**In-Orbit Manufacturing**
CubeSats are being used to test in-space manufacturing concepts, such as 3D printing in microgravity. The goal is to enable on-orbit production of components or structures, reducing launch mass and allowing adaptive, customizable mission architectures.



**Miniaturized X-ray / Gamma-ray Detectors**
Compact high-energy detectors onboard CubeSats support space-based astrophysics, enabling cost-effective observations of cosmic X-ray and gamma-ray sources with small yet powerful instruments.



**Space Debris Removal Technologies**
CubeSats are also platforms for demonstrating debris mitigation concepts, including capture mechanisms and controlled deorbiting systems, contributing to solutions for the growing problem of orbital debris.




SPACE FACULTY
2NDSpace